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Deep Reinforcement Learning for Adaptive Hospital Inventory Optimization

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Abstract — Hospital inventory management has a lot of problems. Some of these problems include running out of stock or having excess stock, items that are already expired and so on. This might happen because outdated system may not be able to keep a track of all items. These systems may also have trouble with changes in demand. They also must deal with rules and regulations in a healthcare system where time is very important.

The aim of our project is to address some of these concerns through developing a particular system. This system provides for forecasting that ensures the proper amount of inventory is always present, preventing any possible shortage or overstock. Using machine learning, deep neural network, and accurate forecasting, the system is aimed at optimizing the inventory and even replenishing it as necessary.

This inventory management system consists of multiple elements working with each other, ensuring the collection of relevant data, generation of forecasts, optimization of stock while maintaining proper fairness. Overall, the development and implementation of this system are revolutionizing the inventory management process at hospitals.

Note to Practitioners— Hospital pharmacy and supply chain teams often struggle with running out of medications or surgical supplies during busy periods. At the same time, they might also have to deal with a lot of extra items on shelves that are about to expire. These items take up space and money. Often go to waste. The problems come from not knowing how many patients will come in suppliers taking amounts of time to deliver and staff having to count items by hand. Our tool uses a type of computer program that learns from how your hospital uses supplies. It helps make decisions about how much to order. Of using a set number of extra supplies or planning with spreadsheets the tool looks at what is in stock, how much has been used recently and how long it takes for suppliers to deliver. Then it suggests how much to the personals must order without having too much. In tests this approach worked well. It kept items available almost all the time, stopped thousands of shortages and saved millions of dollars that was previously spent on extra supplies. It also greatly reduced the number of expired supplies that had to be thrown.

Index Terms—Artificial Intelligence, hospital management, machine learning, patient admission prediction, resource optimization

I. INTRODUCTION

Hospital inventory management is critical for healthcare to work effectively. But it continues to experience numerous challenges. Traditional inventory systems based on tracking or spreadsheets face difficulties due to sudden variations in demand and irregular deliveries as well as complex policies. As a result, hospitals lack medication and equipment or have too many items which get wasted in the end. Another problem associated with this issue is the inability to monitor inventory throughout the facility or multiple facilities.

Such problems were exacerbated by the disruptions caused by pandemics and shortages of various supplies. It becomes clear that there is a need to develop an approach that can manage inventories of hospitals flexibly and efficiently.

Hospital inventory management systems and Artificial Intelligence could combine their efforts to address all of these issues. Indeed, Artificial Intelligence makes use of analytical tools and machine learning to assist in decision making based on the analysis of real-time data taken from the relevant records, sensors, and previous consumption patterns. As a result, there will be a possibility to forecast demand, determine optimal ordering times, and minimize any wastage of resources. Some improvements in this respect have already been made – namely, hospitals gained ability to foresee their needs more accurately and thus reduced overall inventory and ensured proper supply of resources.

The market share of Artificial Intelligence for hospital inventory management will experience significant growth, reaching approximately \$1.78 billion by 2036 compared to only \$571.7 million in 2026. This is due to the benefits provided by Artificial Intelligence in terms of cost-effectiveness and provision of medical services.

In this research, a system architecture is being developed for using Artificial Intelligence in hospital inventory management. The process involves predicting future demands, optimizing inventory management by scheduling the supply chain operations, and monitoring performance indicators of the entire system. Also, there are mechanisms to ensure fairness and security of the entire process.

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II. MACHINE LEARNING ALGORITHM

The following machine learning models will be used for demand prediction in this research for inventory management at hospitals: *Linear Regression*, *Logistic Regression*, *Random Forest*, and *XGBoost*. These models are selected as they can

handle both regression-based continuous demand prediction (e.g., quantity of supplies needed) along with classification tasks (e.g., categorizing items as high-risk for shortage or classifying supply categories for prioritization).

- *Linear Regression* serves as a basic forecasting model that predicts numerical demand. It assumes a linear relationship between previous attributes (like past utilization rate, lead time, and hospital admission) and future inventory requirement. Though it is simple to understand and easy to implement, it works effectively only on data sets having a linear pattern, which may not be applicable in healthcare demand.
- *Logistic Regression* model can be used in applications where there are multiple classes to choose from. For example, predicting whether a particular item is likely to have a stock-out problem or categorizing items like medicines based on their urgency level, such as critical or non-critical.
- *Random Forest* is a type of ensemble algorithm where several trees generated. The predictions are made using all these trees, thereby resulting in high accuracy with minimal chances of overfitting. The algorithm performs very well when dealing with non-linear models and mixed features in hospital-related datasets.
- *XG Boost* (Extreme Gradient Boosting) is mainly used as it is one of the best performing algorithms in structured data sets. This is because the algorithm works by making various decision trees sequentially, which is coupled with the inclusion of regularization to avoid any overfitting.

The models are trained with the help of past hospital inventory data (usage, lead time, and demand variance). Performance evaluation is carried out by MAE, RMSE, R2 (regression), and accuracy/F1 (classification) measures. It was found that XG Boost and random forest give the best results for nonlinear demand curves in the healthcare domain, whereas linear and logistic regression can be used as baseline models.

III. MATH

A. Equations used:

- *Linear regression:*

The predicted value is given by:

$$\hat{y}_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} \quad (1)$$

Where:

$\hat{y}_i$ = predicted demand for observation i

β_0 = intercept

β_j = coefficient of feature j

x_{ij} = value of feature j for observation i

p = number of features

The model minimizes the sum of squared errors (Ordinary Least Squares loss):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

- *Logistic regression:*

Linear Combination:

$$z_i = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} \quad (3)$$

Sigmoid Function:

$$p_i = \frac{1}{1 + e^{-z_i}} \quad (4)$$

Binary Cross-Entropy Loss:

$$L = -\left(\frac{1}{n}\right) \sum [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (5)$$

- *Random Forest:*

Regression Prediction:

$$\hat{y}(x) = \frac{1}{T} \sum_{t=1}^T f_t(x) \quad (6)$$

Where:

T = number of trees

$f_t(x)$ = prediction from tree T

Classification Prediction (Majority Vote):

$$\hat{y} = \text{mode}\{f_1(x), \dots, f_T(x)\} \quad (7)$$

Impurity Measures:

Mean Squared Error (Regression Split Criterion):

$$MSE = \left(\frac{1}{n}\right) \sum_{i=1}^n (y_i - \bar{y})^2 \quad (8)$$

Gini Impurity (Classification):

$$G = 1 - \sum_{k=1}^C p_k^2 \quad (9)$$

Entropy (Classification):

$$H = -\sum p_k \log(p_k) \quad (10)$$

- *XG Boost:*

Additive Prediction Model:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i) \quad (11)$$

Objective Function:

$$\Omega(f) = \gamma T + \left(\frac{1}{2}\right) \lambda \sum_{j=1}^T w_j^2 \quad (12)$$

Split Gain Formula:

$$Gain = \left(\frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] \right) - \gamma \quad (13)$$

B. Algorithms

1. *Linear Regression*: It is a basic form of parametric modelling technique where fitting of a straight line into data points is done to achieve minimal error square. This is employed as a baseline technique to predict continuous values of demand for inventories based on linear relationships with features.

2. *Logistic Regression*: It is a probabilistic modelling technique that uses logistic functions to generalize linear regression models for classification purposes. It can be used to solve classification task such as classifying products as having a high probability of experiencing a stockout situation. It deals with binary outcomes.

3. *Random Forest*: This is a type of ensemble technique where various decision trees are created using datasets obtained through bootstrap sampling process. Random variables are then selected when deciding on what splits to use. Predictions are made by either taking an average or mode value, depending on regression or classification problem respectively.

4. *XGBoost* (Extreme Gradient Boosting): It refers to a scalable and regularized form of gradient boosting algorithm that constructs decision trees sequentially to minimize loss and penalize model complexity. This is highly suitable for structured data tasks as it performs best in most applications.

IV. WORKING PIPELINE OF THE MODEL

The pipeline processes hospital inventory data (e.g., consumption of medications, consumables, surgical supplies) in a sequential, automated flow:

A. Data Generation:

The artificial records of hospital inventories are created to emulate 1-2 years of the actual daily or weekly usage by different departments and various types of items. The emulation is done considering the initial demand, which can be based on probabilistic models (e.g. Poisson distribution for normal use) along with seasonal fluctuations (higher demand for medication for infections during monsoons) and emergencies represented by high impact random events. Information about batches, including their shelf-life and expiration periods, is also included. This process allows enough variety and data, while avoiding any privacy concerns or lack of data from hospitals.

B. Pre Processing:

The generated raw data is processed systematically to ensure consistency in analysis and modeling. Missing data is processed by mean or median imputation in the case of numeric data, whereas mode imputation is applied in the case of categorical data. Important missing data can be eliminated from the dataset. The date columns in the dataset are converted into the datetime format so that the time-based

features can be derived from the data. The outliers in the demand and stock features are identified by the interquartile range or Z-score method and corrected to avoid distorting the model.

C. Feature Engineering:

Sophisticated temporal and operational features are created to understand the behavior of demand and inventories. The use of rolling averages (7-days and 14-days) creates a feature that understands the short-run trends of consumer demands, while lagging variables consider the previous period's demands for time-series dependency. Features related to seasonality include the month of the year and the week of the year. The velocity of consumption is determined by taking the ratio of the average daily consumption to the total stock on hand. The probability of lead-time risk involves the probability of supplier delays and the level of the inventory.

D. Demand Forecast:

An XG Boost Regressor is used to predict the future 14-day demand with gradient boosting and regularization. The model creates decision trees one after another, each correcting the errors made by the previous tree, which allows it to model complicated nonlinear relationships among seasonality, inventory, emergencies, and consumption. Hyperparameter tuning is carried out through cross-validation to avoid overfitting. Importance values for each feature are calculated to understand the factors affecting demand. Performance of the model is assessed based on RMSE and MAE. The predictions generated help in planning procurement in advance to avoid shortages or excess stocks.

E. Stock Out Risk:

The Random Forest Classifier predicts the probability of stockouts in the next seven days. The classifier creates numerous decision trees through bootstrap sampling and randomly selected predictors to minimize variance. Features used by the classifier comprise predicted demand, available inventory, lead times, stock coverages, and demand volatility. Each tree generates a class label, and the outcome is obtained via majority voting or average probability values. Performance measures include accuracy, precision-recall curve, F1 score, and area under the receiver operating characteristic curve (ROC). Potential threats prompt an alert for early procurement interventions. This risk probability estimation facilitates preventive measures instead of reactive stock replenishment post-stockouts.

F. Expiry Risk:

The Logistic Regression algorithm is used in batch form to calculate the probability of expiration of a drug batch prior to its usage. The logistic regression algorithm relies on the predictor variables of the time left for the expiration date, batch age, expected demand, stock availability, and rate of consumption. With the use of the sigmoid function, it transforms the linear combination of the predictors to a probability score that can be interpreted by the end-user. Its model parameters give insights about the predictors that have an impact on the expiry probability of a particular batch.

G. Cost Impact:

Financials in a simulation module determine the financial consequences associated with risk predictions. Stockout cost is calculated based on the number of delayed surgeries multiplied by the opportunity cost per procedure. Overstock cost is determined by multiplying the holding cost percentage against the value of excess stock. Expired waste is determined by multiplying the number of expiries by the unit price. The summation of all these costs helps determine the overall loss that could be avoided. Scenario planning enables changes to reorder points, delivery times, and safety stocks to assess their effect on costs.

H. Dashboard and Reporting:

All predictive output and cost calculations are included in an interactive Streamlit dashboard connected to the API. This includes displaying the forecast, stockout probability heatmap, expiration alerts, and cost implications all in one view. It is possible to filter results by department, type of item, and time. Users can do “what if” analysis to see the potential outcome of certain changes to inventory control and then produce reports that can be reviewed by hospital management. The dashboard layout will provide easy navigation for pharmacists, supply chain management professionals, and hospital administrators to gain insight into inventory control.

V. CONCLUSION

Hospital inventory management is an essential but continually problematic area within health care operations systems, in which the use of traditional methods involving manual inventory tracking, spreadsheets, and rules-based inventory replenishment usually leads to critical out-of-stocks, excessive overstocks, massive amounts of waste due to expiration of items, and very high expenses. Such problems arise from fluctuating demand patterns, variable lead time periods, occasional emergencies, seasonality, and lack of visibility within the entire multi-department/multi-facility organization, affecting both patients' safety and efficient resource management.

The present paper describes an AI solution for intelligent hospital inventory management using four different supervised machine learning models namely, Linear Regression, Logistic Regression, Random Forest, and XGBoost to ensure accurate forecasting of demand levels and shortages. The linear and logistic regression models serve as baseline approaches, whereas Random Forest and XGBoost allow capturing complex relationships in the healthcare industry.

With the help of the predictive analytics tools mentioned on the dataset that includes past usage rates, lead times, admissions, and other related information, reliable predictions can be generated. By bridging foundational automation research with practical healthcare needs, this work contributes to more resilient, sustainable supply chains and lays the groundwork for future integration of real-time IoT and advanced reinforcement learning.

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